ANSWER KEY

빠른정답

01	$\frac{6az^2}{3a^2x^2y^3z}$, $\frac{xy^4}{3a^2x^2y^3z}$			**02**	$\frac{(x-2)^2}{(x-2)(x+1)(x-5)}$, $\frac{(x+5)(x-5)}{(x-2)(x+1)(x-5)}$		
03	$\frac{x+4}{x-6}$	**04**	$\frac{3x-7}{(x-3)(x-1)}$	**05**	$\frac{-2x-1}{(x-3)(x-1)}$	**06**	$\frac{x+3}{(x-3)(x+1)}$
07	$\frac{1}{(x-3)^2}$	**08**	$\frac{-4x+10}{(x+2)(x-4)}$	**09**	$-\frac{x+10}{(x+3)(x-4)}$	**10**	①
11	$\frac{6(x^2-3x+4)}{(x-5)(x-2)(x-1)(x+2)}$			**12**	$\frac{20}{29}$	**13**	$\frac{x+2}{x-1}$
14	②	**15**	③	**16**	④	**17**	$\frac{2}{x(x+4)}$
18	$\frac{5}{(x+3)(x-2)}$	**19**	$\frac{x-2}{(x+1)(x-3)}$	**20**	$\frac{1}{x+4}$	**21**	9
22	4	**23**	①	**24**	③	**25**	6
26	②	**27**	②	**28**	4	**29**	3
30	2	**31**	④	**32**	③	**33**	10
34	②	**35**	②				

오답 메모 · 다시 볼 문항

해설 SOLUTIONS

01 정답 $\frac{6az^2}{3a^2x^2y^3z}$, $\frac{xy^4}{3a^2x^2y^3z}$

$\frac{2z}{ax^2y^3}$, $\frac{y}{3a^2xz}$ 의 분모의 최소공배수가 $3a^2x^2y^3z$ 이므로

두 식을 통분하면 $\frac{6az^2}{3a^2x^2y^3z}$, $\frac{xy^4}{3a^2x^2y^3z}$

02 정답 $\frac{(x-2)^2}{(x-2)(x+1)(x-5)}$, $\frac{(x+5)(x-5)}{(x-2)(x+1)(x-5)}$

$x^2-4x-5=(x-5)(x+1)$,

$x^2-x-2=(x+1)(x-2)$ 이므로

두 식을 통분하면

$\frac{(x-2)^2}{(x-2)(x+1)(x-5)}$, $\frac{(x+5)(x-5)}{(x-2)(x+1)(x-5)}$

03 정답 $\frac{x+4}{x-6}$

$$\frac{x^2+6x+8}{x^2-4x-12}=\frac{(x+2)(x+4)}{(x+2)(x-6)}=\frac{x+4}{x-6}$$

04 정답 $\frac{3x-7}{(x-3)(x-1)}$

$$\frac{1}{x-3}+\frac{2}{x-1}=\frac{x-1+2(x-3)}{(x-3)(x-1)}=\frac{3x-7}{(x-3)(x-1)}$$

05 정답 $\frac{-2x-1}{(x-3)(x-1)}$

$$\frac{x-4}{x^2-4x+3}-\frac{3}{x-3}=\frac{x-4}{(x-3)(x-1)}-\frac{3}{x-3}$$
$$=\frac{x-4}{(x-3)(x-1)}-\frac{3(x-1)}{(x-3)(x-1)}$$
$$=\frac{-2x-1}{(x-3)(x-1)}$$

06 정답 $\frac{x+3}{(x-3)(x+1)}$

$$\frac{x^2+x-6}{x^2+2x-15}\times\frac{x+5}{x^2-x-2}$$
$$=\frac{(x+3)(x-2)}{(x-3)(x+5)}\times\frac{x+5}{(x+1)(x-2)}$$
$$=\frac{x+3}{(x-3)(x+1)}$$

07 정답 $\frac{1}{(x-3)^2}$

$$\frac{x+4}{x^2-9}\div\frac{x^2+x-12}{x+3}$$
$$=\frac{x+4}{(x+3)(x-3)}\div\frac{(x+4)(x-3)}{x+3}$$
$$=\frac{x+4}{(x+3)(x-3)}\times\frac{x+3}{(x+4)(x-3)}$$
$$=\frac{1}{(x-3)^2}$$

08 정답 $\frac{-4x+10}{(x+2)(x-4)}$

$$\frac{x-1}{x+2}-\frac{x-3}{x-4}=\frac{(x-1)(x-4)-(x-3)(x+2)}{(x+2)(x-4)}$$
$$=\frac{-4x+10}{(x+2)(x-4)}$$

09 정답 $-\frac{x+10}{(x+3)(x-4)}$

$x^2+5x+7=(x+2)(x+3)+1$

$x^2-2x-6=(x+2)(x-4)+2$

$$\therefore\ \frac{x^2+5x+7}{x+3}-\frac{x^2-2x-6}{x-4}$$
$$=\frac{(x+2)(x+3)+1}{x+3}-\frac{(x+2)(x-4)+2}{x-4}$$
$$=x+2+\frac{1}{x+3}-(x+2)-\frac{2}{x-4}$$
$$=\frac{1}{x+3}-\frac{2}{x-4}$$
$$=\frac{x-4-2(x+3)}{(x+3)(x-4)}$$
$$=-\frac{x+10}{(x+3)(x-4)}$$

10 정답 ①

$$\frac{x+2}{x^2+2x}-\frac{1}{x-2}=\frac{x+2}{x(x+2)}-\frac{1}{x-2}$$
$$=\frac{1}{x}-\frac{1}{x-2}$$
$$=-\frac{2}{x(x-2)}$$

해설 SOLUTIONS

11 정답 $\frac{6(x^2-3x+4)}{(x-5)(x-2)(x-1)(x+2)}$

$$\frac{1}{x-5}+\frac{1}{x-1}-\frac{1}{x-2}-\frac{1}{x+2}$$
$$=\left(\frac{1}{x-5}-\frac{1}{x-2}\right)+\left(\frac{1}{x-1}-\frac{1}{x+2}\right)$$
$$=\frac{x-2-(x-5)}{(x-5)(x-2)}+\frac{x+2-(x-1)}{(x-1)(x+2)}$$
$$=\frac{3}{(x-5)(x-2)}+\frac{3}{(x-1)(x+2)}$$
$$=\frac{3(x-1)(x+2)+3(x-5)(x-2)}{(x-5)(x-2)(x-1)(x+2)}$$
$$=\frac{3x^2+3x-6+3x^2-21x+30}{(x-5)(x-2)(x-1)(x+2)}$$
$$=\frac{6x^2-18x+24}{(x-5)(x-2)(x-1)(x+2)}$$
$$=\frac{6(x^2-3x+4)}{(x-5)(x-2)(x-1)(x+2)}$$

12 정답 $\frac{20}{29}$

$$\frac{a+b}{a-b}-\frac{a-b}{a+b}=\frac{(a+b)^2-(a-b)^2}{(a-b)(a+b)}=\frac{4ab}{(a-b)(a+b)}$$
$$\frac{a+b}{a-b}+\frac{a-b}{a+b}=\frac{(a+b)^2+(a-b)^2}{(a-b)(a+b)}=\frac{2(a^2+b^2)}{(a-b)(a+b)}$$
$$\therefore\left(\frac{a+b}{a-b}-\frac{a-b}{a+b}\right)\div\left(\frac{a+b}{a-b}+\frac{a-b}{a+b}\right)$$
$$=\frac{4ab}{(a-b)(a+b)}\times\frac{(a-b)(a+b)}{2(a^2+b^2)}$$
$$=\frac{2ab}{a^2+b^2}=\frac{20}{29}$$

13 정답 $\frac{x+2}{x-1}$

$$A\div\frac{x^3+8}{x^4+4x^2+16}\times\frac{x^2-x}{x^3-8}$$
$$=A\times\frac{(x^2-2x+4)(x^2+2x+4)}{(x+2)(x^2-2x+4)}$$
$$\times\frac{x(x-1)}{(x-2)(x^2+2x+4)}$$
$$=A\times\frac{x(x-1)}{(x+2)(x-2)}$$

따라서 $A\times\frac{x(x-1)}{(x+2)(x-2)}=\frac{x}{x-2}$ 이므로

$$A=\frac{x}{x-2}\times\frac{(x+2)(x-2)}{x(x-1)}=\frac{x+2}{x-1}$$

14 정답 ②

$$\frac{1}{a-1}-\frac{1}{a+1}-\frac{2}{a^2+1}-\frac{4}{a^4+1}$$
$$=\frac{(a+1)-(a-1)}{(a-1)(a+1)}-\frac{2}{a^2+1}-\frac{4}{a^4+1}$$
$$=\frac{2}{a^2-1}-\frac{2}{a^2+1}-\frac{4}{a^4+1}$$
$$=\frac{2(a^2+1)-2(a^2-1)}{(a^2-1)(a^2+1)}-\frac{4}{a^4+1}$$
$$=\frac{4}{a^4-1}-\frac{4}{a^4+1}$$
$$=\frac{4(a^4+1)-4(a^4-1)}{a^8-1}$$
$$=\frac{2\times4}{a^8-1}=\frac{2^3}{a^8-1}$$

$p=3,\ q=8$ 이므로 $p+q=11$ 이다.

15 정답 ③

$\frac{3x+23}{x^2-3x-4}=\frac{a}{x+1}+\frac{b}{x-4}$ 에서

$$\frac{3x+23}{x^2-3x-4}=\frac{a(x-4)+b(x+1)}{(x+1)(x-4)}$$
$$\frac{3x+23}{x^2-3x-4}=\frac{(a+b)x-4a+b}{x^2-3x-4}$$

이때 주어진 등식은 $x\neq-1,\ x\neq4$ 인 모든 실수 x 에 대하여 성립하므로

$a+b=3,\ -4a+b=23$

$\therefore\ a=-4,\ b=7$

$\therefore\ ab=-28$

16 정답 ④

$12x^2-2ax+b\neq0$ 인 모든 실수 x 에 대하여 유리식 $\frac{3x^2+2x+a}{12x^2-2ax+b}$ 의 값이 항상 일정하므로

$\frac{\alpha\times A(x)}{\beta\times A(x)}$ ($\alpha,\ \beta$ 는 상수이고 $\beta\neq0,\ A(x)$ 는 x 에 대한 다항식) 꼴이어야 한다.

즉, $\frac{3x^2+2x+a}{12x^2-2ax+b}=\frac{3x^2+2x+a}{4\left(3x^2-\frac{a}{2}x+\frac{b}{4}\right)}$ 이므로

$-\frac{a}{2}=2,\ \frac{b}{4}=a$

$\therefore\ a=-4,\ b=-16$

$\therefore\ a-b=12$

해설 SOLUTIONS

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정답 $\dfrac{2}{x(x+4)}$

$$\frac{1}{x(x+2)}+\frac{1}{(x+2)(x+4)}$$
$$=\frac{1}{(x+2)-x}\left(\frac{1}{x}-\frac{1}{x+2}\right)+\frac{1}{(x+4)-(x+2)}\left(\frac{1}{x+2}-\frac{1}{x+4}\right)$$
$$=\frac{1}{2}\left(\frac{1}{x}-\frac{1}{x+4}\right)$$
$$=\frac{(x+4)-x}{2x(x+4)}=\frac{2}{x(x+4)}$$

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정답 $\dfrac{5}{(x+3)(x-2)}$

$$\frac{4}{(x-2)(x+2)}+\frac{1}{(x+2)(x+3)}$$
$$=\frac{1}{(x+2)-(x-2)}\left(\frac{4}{x-2}-\frac{4}{x+2}\right)+\frac{1}{(x+3)-(x+2)}\left(\frac{1}{x+2}-\frac{1}{x+3}\right)$$
$$=\frac{1}{x-2}-\frac{1}{x+3}$$
$$=\frac{(x+3)-(x-2)}{(x+3)(x-2)}$$
$$=\frac{5}{(x+3)(x-2)}$$

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정답 $\dfrac{x-2}{(x+1)(x-3)}$

$$\frac{\dfrac{1}{x+1}}{\dfrac{x-3}{x-2}}=\frac{1}{x+1}\times\frac{x-2}{x-3}=\frac{x-2}{(x+1)(x-3)}$$

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정답 $\dfrac{1}{x+4}$

$$\frac{\dfrac{1}{x+2}}{1+\dfrac{2}{x+2}}$$
$$=\frac{1}{x+2}\times\frac{1}{1+\dfrac{2}{x+2}}$$
$$=\frac{1}{x+2}\times\frac{x+2}{x+4}=\frac{1}{x+4}$$

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정답 9

$$\frac{2}{(x-5)(x-3)}+\frac{3}{x(x-3)}+\frac{2}{x(x+2)}$$
$$=\left(\frac{1}{x-5}-\frac{1}{x-3}\right)+\left(\frac{1}{x-3}-\frac{1}{x}\right)+\left(\frac{1}{x}-\frac{1}{x+2}\right)$$
$$=\frac{1}{x-5}-\frac{1}{x+2}=\frac{7}{(x-5)(x+2)}$$

따라서 $\dfrac{7}{(x-5)(x+2)}=\dfrac{a}{(x-5)(x+b)}$ 이고

이 식이 x 에 대한 항등식이므로

$a=7,\ b=2$

$\therefore\ a+b=9$

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정답 4

$$\frac{x}{x-2}-\frac{x+1}{x-1}-\frac{x+2}{x}+\frac{x+3}{x+1}$$
$$=\frac{(x-2)+2}{x-2}-\frac{(x-1)+2}{x-1}-\frac{x+2}{x}+\frac{(x+1)+2}{x+1}$$
$$=\left(1+\frac{2}{x-2}\right)-\left(1+\frac{2}{x-1}\right)-\left(1+\frac{2}{x}\right)+\left(1+\frac{2}{x+1}\right)$$
$$=\frac{2}{x-2}-\frac{2}{x-1}-\frac{2}{x}+\frac{2}{x+1}$$
$$=\frac{2}{(x-2)(x-1)}-\frac{2}{x(x+1)}$$
$$=\frac{2\{x(x+1)-(x-2)(x-1)\}}{x(x-2)(x-1)(x+1)}$$
$$=\frac{8x-4}{x(x-2)(x-1)(x+1)}$$

즉,

$$\frac{8x-4}{x(x-2)(x-1)(x+1)}=\frac{ax+b}{x(x-2)(x-1)(x+1)}$$

가 x 에 대한 항등식이므로 $a=8,\ b=-4$

$\therefore\ a+b=4$

해설 SOLUTIONS

23 정답 ①

$$\frac{3x^2-3x+5}{x-1}-\frac{3x^2+9x-10}{x+3}$$
$$=\frac{3x(x-1)+5}{x-1}-\frac{3x(x+3)-10}{x+3}$$
$$=3x+\frac{5}{x-1}-\left(3x-\frac{10}{x+3}\right)$$
$$=\frac{5}{x-1}+\frac{10}{x+3}$$
$$=\frac{5(x+3)+10(x-1)}{(x-1)(x+3)}$$
$$=\frac{15x+5}{(x-1)(x+3)}$$

따라서 $f(x)=15x+5$ 이므로 $f(1)=20$ 이다.

24 정답 ③

$$\frac{1}{4}+\frac{1}{28}+\frac{1}{70}+\cdots+\frac{1}{550}$$
$$=\frac{1}{1\times4}+\frac{1}{4\times7}+\frac{1}{7\times10}+\cdots+\frac{1}{22\times25}$$
$$=\frac{1}{3}\left\{\left(\frac{1}{1}-\frac{1}{4}\right)+\left(\frac{1}{4}-\frac{1}{7}\right)+\cdots+\left(\frac{1}{22}-\frac{1}{25}\right)\right\}$$
$$=\frac{1}{3}\left(1-\frac{1}{25}\right)=\frac{8}{25}$$

25 정답 6

$$\frac{1}{x^2+9x+18}+\frac{1}{x^2+3x}+\frac{1}{x^2-3x}$$
$$=\frac{1}{(x+6)(x+3)}+\frac{1}{x(x+3)}+\frac{1}{x(x-3)}$$
$$=\frac{1}{3}\left\{\left(\frac{1}{x+3}-\frac{1}{x+6}\right)+\left(\frac{1}{x}-\frac{1}{x+3}\right)+\left(\frac{1}{x-3}-\frac{1}{x}\right)\right\}$$
$$=\frac{1}{3}\left(\frac{1}{x-3}-\frac{1}{x+6}\right)$$
$$=\frac{3}{(x+6)(x-3)}$$

따라서 $\frac{3}{(x+6)(x-3)}=\frac{1}{12}$ 이므로 $(x+6)(x-3)=36$

$x^2+3x-54=0,\ (x+9)(x-6)=0$

$\therefore\ x=6\ (\because\ x>0)$

26 정답 ②

$$\frac{4}{(x+3)(x+7)}+\frac{5}{(x+7)(x+12)}+\frac{6}{(x+12)(x+18)}+\cdots+\frac{9}{(x+33)(x+42)}$$
$$=\left(\frac{1}{x+3}-\frac{1}{x+7}\right)+\left(\frac{1}{x+7}-\frac{1}{x+12}\right)+\left(\frac{1}{x+12}-\frac{1}{x+18}\right)+\cdots+\left(\frac{1}{x+33}-\frac{1}{x+42}\right)$$
$$=\frac{1}{x+3}-\frac{1}{x+42}$$
$$=\frac{1}{6}-\frac{1}{45}=\frac{13}{90}$$

27 정답 ②

$$\frac{1}{1+\dfrac{1}{1+\dfrac{1}{x}}}=\frac{1}{1+\dfrac{1}{\dfrac{x+1}{x}}}$$
$$=\frac{1}{1+\dfrac{x}{x+1}}$$
$$=\frac{1}{\dfrac{(x+1)+x}{x+1}}$$
$$=\frac{x+1}{2x+1}$$

즉, $\frac{x+1}{2x+1}=\frac{ax+b}{cx+1}$ 는 x 에 대한 항등식이므로

$a=1,\ b=1,\ c=2$

$\therefore\ a+2b+3c=1+2+6=9$

28 정답 4

$$f(x)=6-\frac{2+\dfrac{4}{x-2}}{2-\dfrac{4}{x+2}}$$
$$=6-\frac{\dfrac{2x}{x-2}}{\dfrac{2x}{x+2}}$$
$$=6-\frac{x+2}{x-2}=\frac{5x-14}{x-2}$$

따라서 $f(a)=3$ 에서 $\frac{5a-14}{a-2}=3$

$5a-14=3a-6\ \therefore\ a=4$

해설 SOLUTIONS

29 정답 3

$\frac{x}{4}=\frac{y}{5}=k\ (k\neq 0)$ 이라 하면 $x=4k,\ y=5k$

$\therefore\ \frac{8x+2y}{x+2y}=\frac{32k+10k}{4k+10k}=\frac{42k}{14k}=3$

30 정답 2

$x:y=3:2$ 이므로 $\frac{x}{3}=\frac{y}{2}$

$\frac{x}{3}=\frac{y}{2}=k\ (k\neq 0)$ 이라 하면 $x=3k,\ y=2k$

$\therefore\ \frac{2x^2+4xy-4y^2}{x^2+y^2}=\frac{18k^2+24k^2-16k^2}{9k^2+4k^2}=\frac{26k^2}{13k^2}=2$

31 정답 ④

$a:b=3:5$ 이므로 $a=3k,\ b=5k\ (k\neq 0)$ 이라 하면

$\frac{3b^2-2ab}{4a^2+ab}=\frac{3\times(5k)^2-2\times 3k\times 5k}{4\times(3k)^2+3k\times 5k}=\frac{45k^2}{51k^2}=\frac{15}{17}$

32 정답 ③

$x^2-5x-1=0$ 에서 $x\neq 0$ 이므로 양변을 x 로 나누면

$x-\frac{1}{x}=5$

$\therefore\ 2x^2-3x+6+\frac{3}{x}+\frac{2}{x^2}$

$=2\left(x^2+\frac{1}{x^2}\right)-3\left(x-\frac{1}{x}\right)+6$

$=2\left\{\left(x-\frac{1}{x}\right)^2+2\right\}-3\left(x-\frac{1}{x}\right)+6$

$=2\times 27-3\times 5+6$

$=45$

33 정답 10

$\frac{x+y}{4}=y+z=\frac{z+x}{7}=k\ (k\neq 0)$ 이라 하면

$x+y=4k,\ y+z=k,\ z+x=7k$ ····· ㉠

위의 세 식을 더하면

$2(x+y+z)=12k\ \therefore\ x+y+z=6k$ ····· ㉡

㉠, ㉡에 의하여

$x=5k,\ y=-k,\ z=2k$

$\therefore\ \frac{x^2+y^2+z^2}{xy+yz+zx}$

$=\frac{(5k)^2+(-k)^2+(2k)^2}{5k\times(-k)+(-k)\times 2k+2k\times 5k}=\frac{30k^2}{3k^2}=10$

34 정답 ②

$a+b-c=0$ 에서

$a+b=c,\ b-c=-a,\ a-c=-b$

이므로

$\frac{a^2+b^2}{ab}-\frac{b^2+c^2}{bc}-\frac{a^2+c^2}{ac}=\frac{a}{b}+\frac{b}{a}-\frac{b}{c}-\frac{c}{b}-\frac{a}{c}-\frac{c}{a}$

$=\frac{b-c}{a}+\frac{a-c}{b}-\frac{a+b}{c}$

$=\frac{-a}{a}+\frac{-b}{b}-\frac{c}{c}$

$=-1+(-1)+(-1)=-3$

35 정답 ②

$x+y-z=0$ ····· ㉠

$2x-y+2z=0$ ····· ㉡

㉠+㉡을 하면

$3x+z=0\ \therefore\ z=-3x$

$z=-3x$ 를 ㉠에 대입하면

$x+y-(-3x)=0\ \therefore\ y=-4x$

$\therefore\ \frac{z^2+xy}{xy+yz-zx}$

$=\frac{(-3x)^2+x\times(-4x)}{x\times(-4x)+(-4x)\times(-3x)-(-3x)\times x}$

$=\frac{5x^2}{11x^2}=\frac{5}{11}$